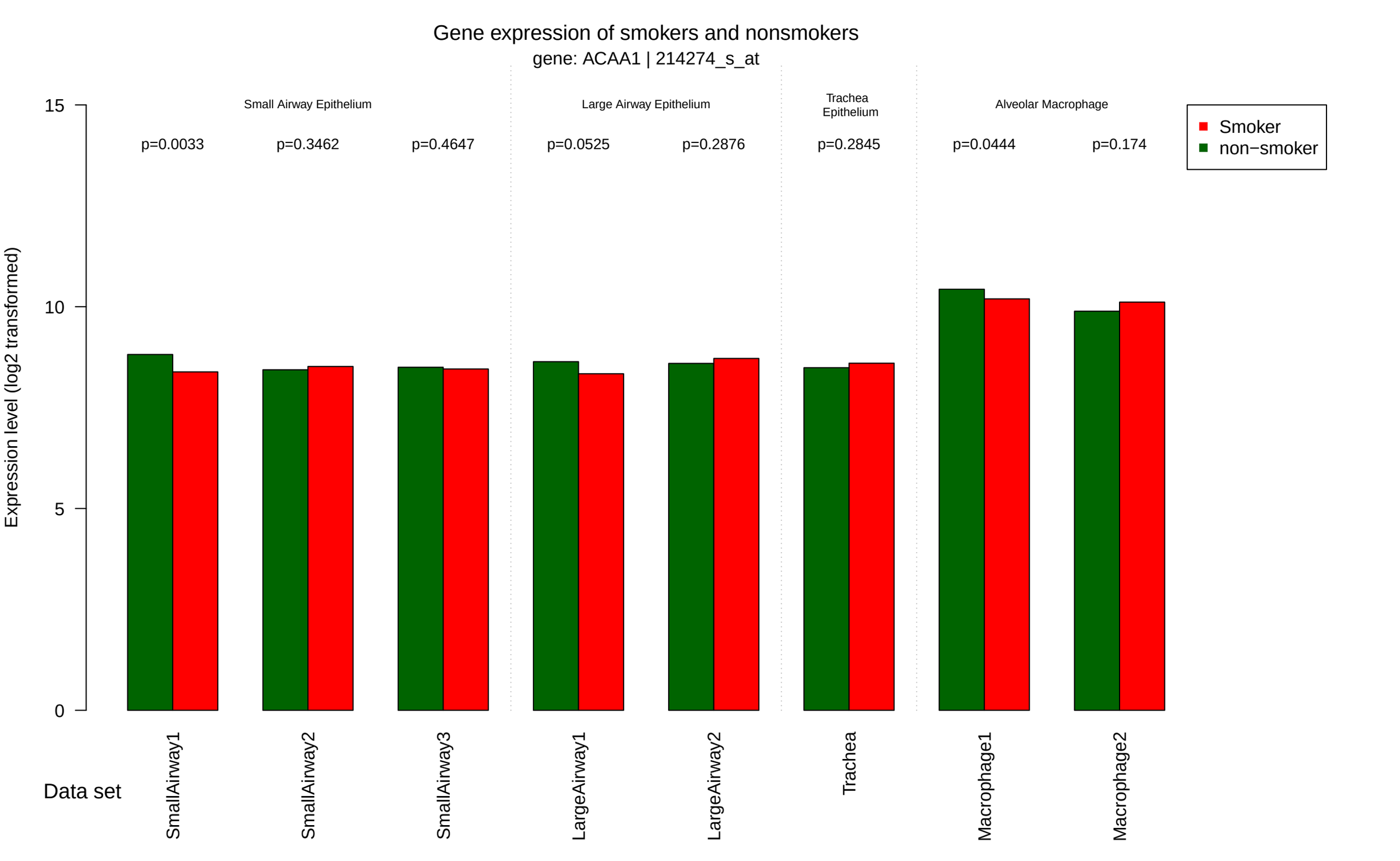
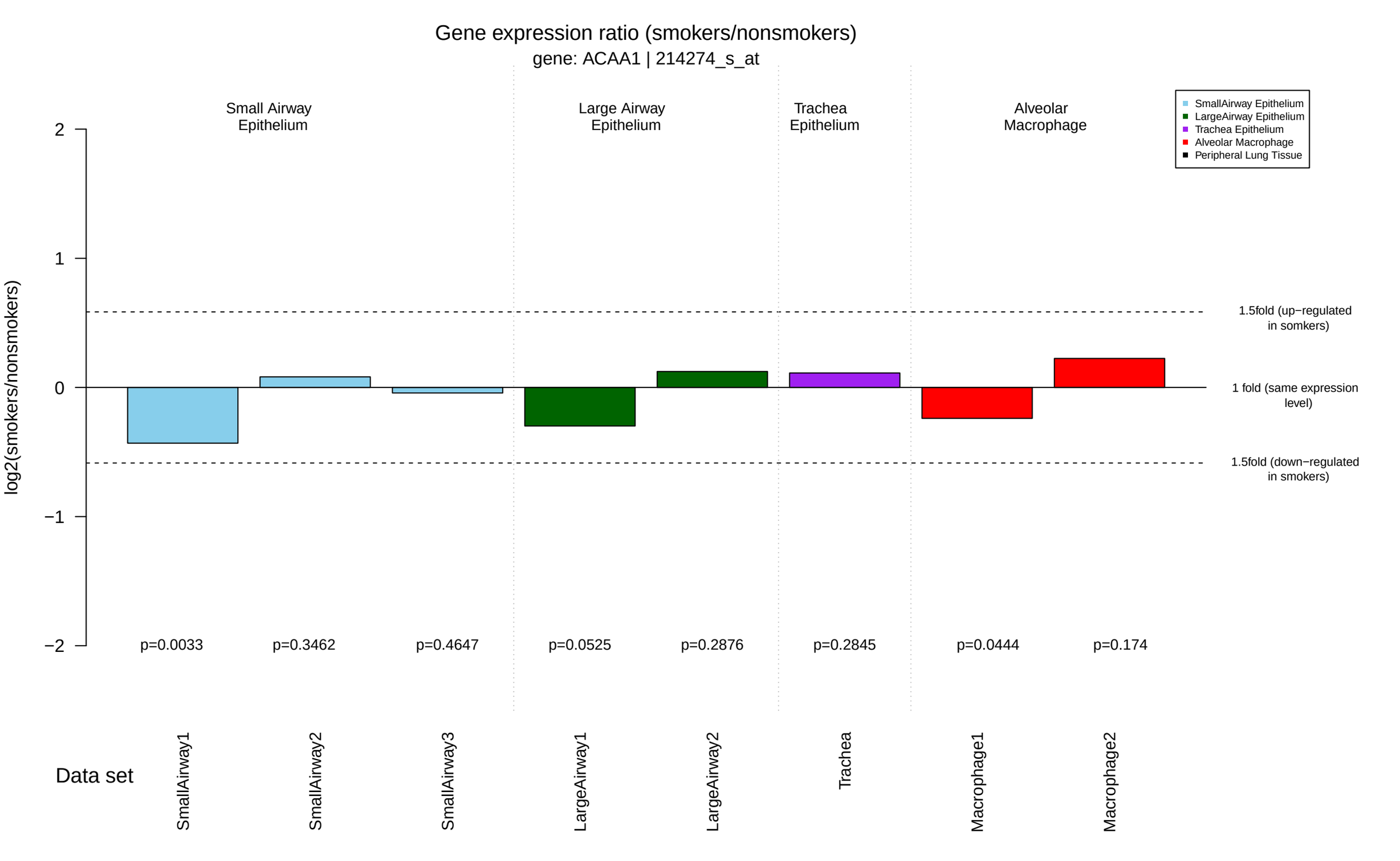
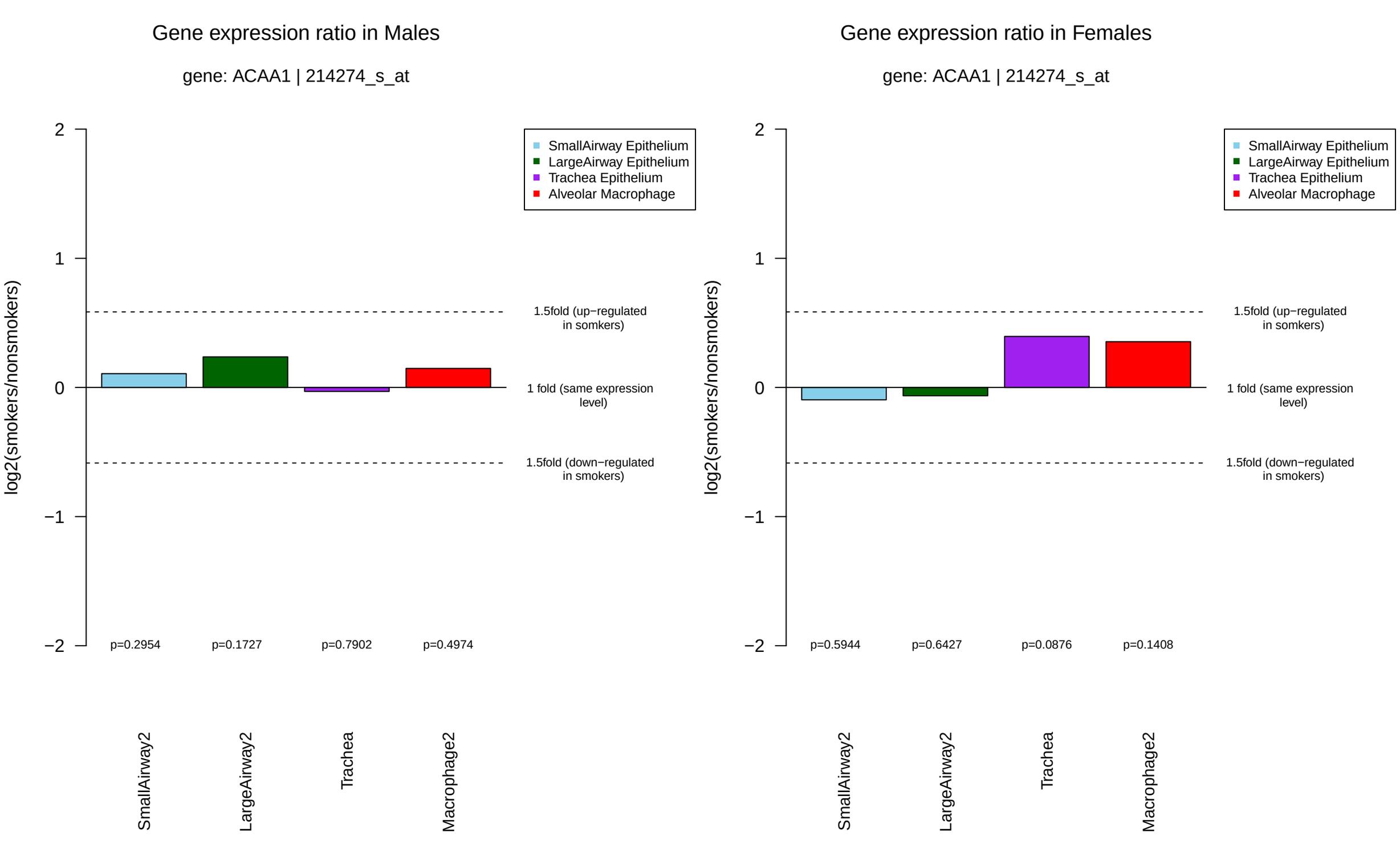
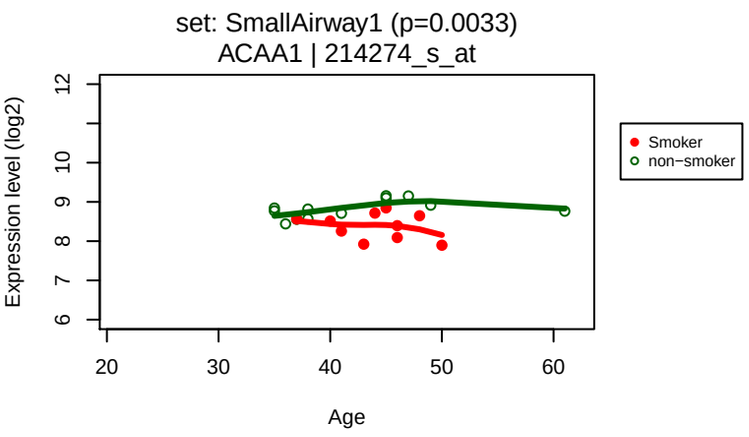
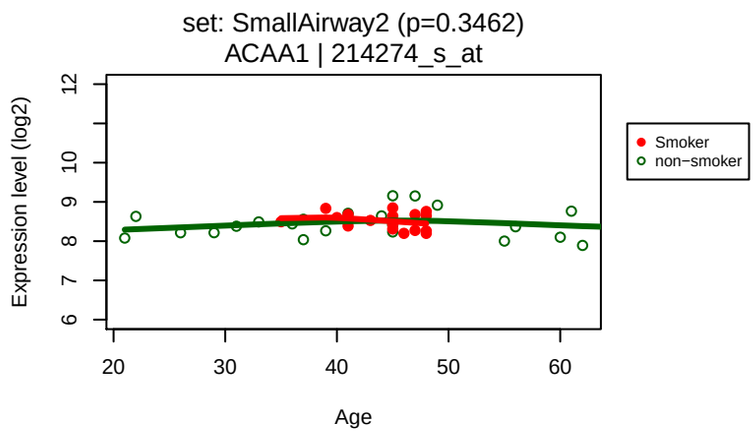
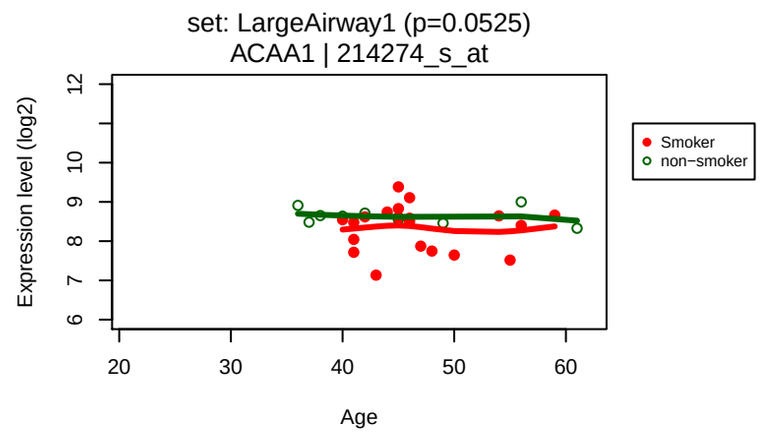

Gene expression of smokers and nonsmokers
gene: ACAA1 | 214274_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0033
p=0.3462
p=0.4647
p=0.0525
p=0.2876
p=0.2845
p=0.0444
p=0.174
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

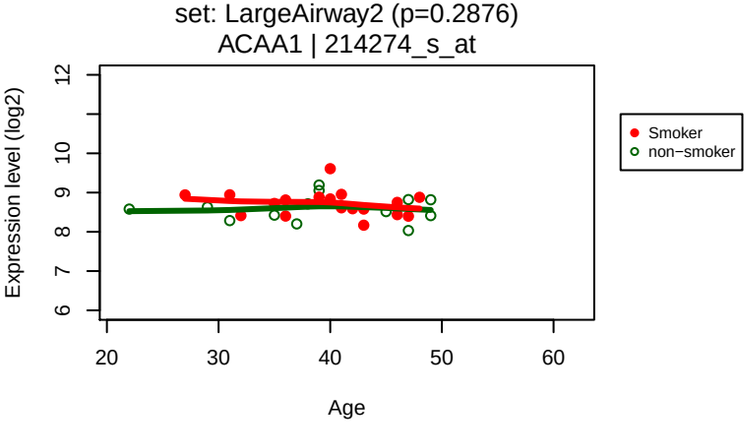

Gene expression ratio (smokers/nonsmokers)
gene: ACAA1 | 214274_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0033
p=0.3462
p=0.4647
p=0.0525
p=0.2876
p=0.2845
p=0.0444
p=0.174
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

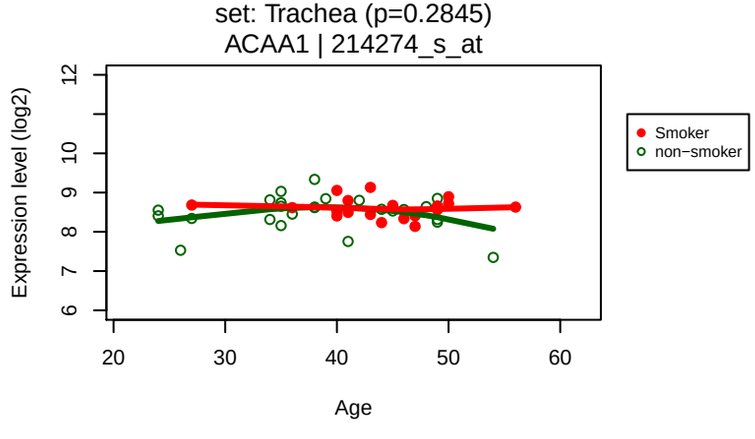

Gene expression ratio in Males
gene: ACAA1 | 214274_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2954
p=0.1727
p=0.7902
p=0.4974
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACAA1 | 214274_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5944
p=0.6427
p=0.0876
p=0.1408
SmallAirway2
LargeAirway2
Trachea
Macrophage2

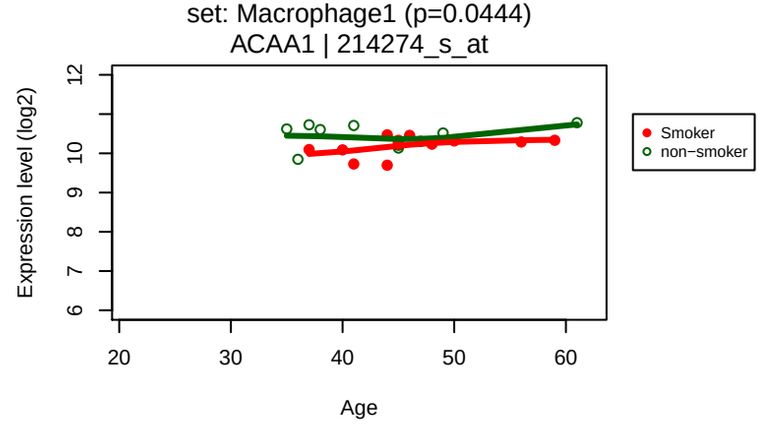

set: SmallAirway1 (p=0.0033)
ACAA1 | 214274_s_at
Expression level (log2)
Age
Smoker
non−smoker

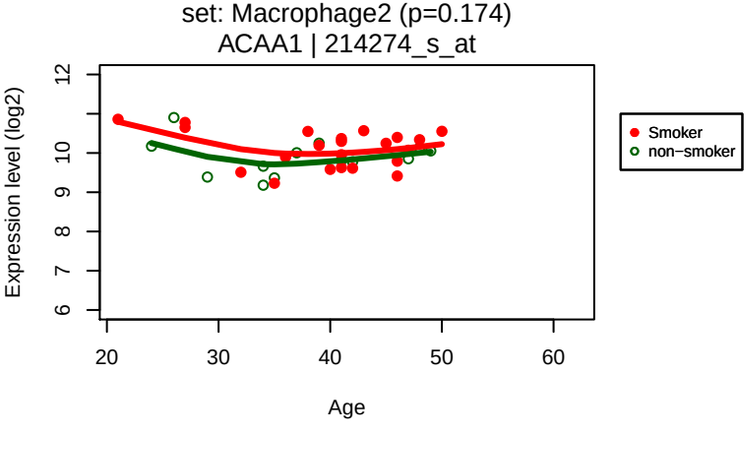

set: SmallAirway2 (p=0.3462)
ACAA1 | 214274_s_at
Expression level (log2)
Age
Smoker
non−smoker

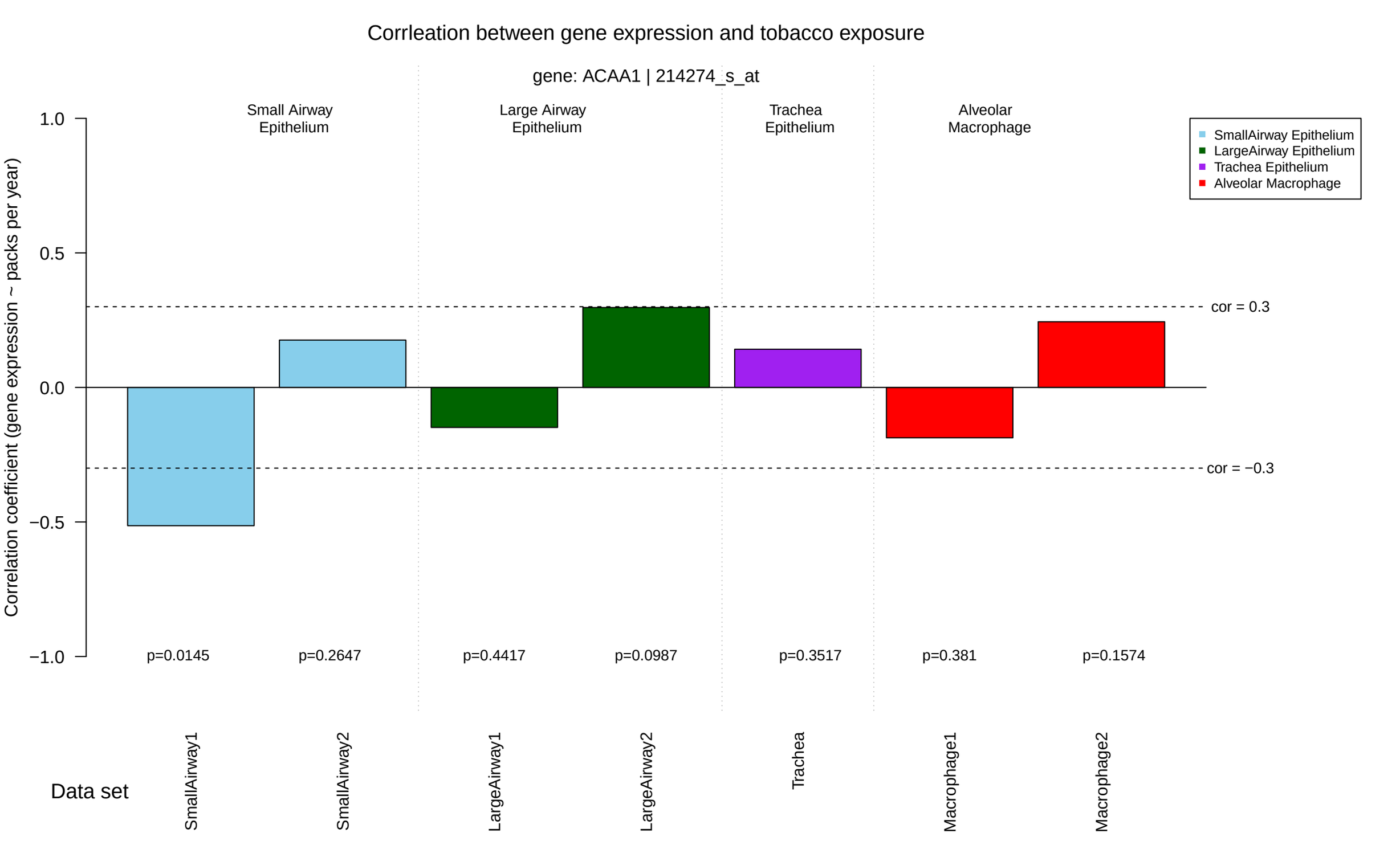

set: LargeAirway1 (p=0.0525)
ACAA1 | 214274_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2876)
ACAA1 | 214274_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2845)
ACAA1 | 214274_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0444)
ACAA1 | 214274_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.174)
ACAA1 | 214274_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACAA1 | 214274_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0145
p=0.2647
p=0.4417
p=0.0987
p=0.3517
p=0.381
p=0.1574
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2